

Defining Relapse and Disease Activity in Neurosarcoidosis

A Systematic Review and Proposed Framework

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Abstract

Purpose of Review

There is no consensus or standardized framework to characterize disease activity in CNS neurosarcoidosis, and the existing literature is highly heterogeneous. Here, we aimed to aggregate all longitudinal monitoring metrics in neurosarcoidosis through a systematic review. The review protocol is registered in PROSPERO (ID:533857). Articles were included if they incorporated at least 1 longitudinal monitoring or outcome metric relevant to neurosarcoidosis disease activity. In addition, we aimed to establish an evidence-based clinical framework for characterizing, monitoring, and communicating neurosarcoidosis disease activity, applicable to both clinical care and research.

Recent Findings

Of 387 articles initially identified, 67 met the inclusion criteria. Most studies focused on metrics within a single domain (clinical, imaging, or laboratory) or analyzed multiple domains independently. Often, the definitions were not formally outlined. Only 11 studies included a multidomain monitoring metric. Our proposed framework integrates 3 critical domains for assessing the neurosarcoidosis disease activity: (1) clinical, (2) imaging, and (3) laboratory and supports a standardized characterization of the disease, especially when domain results are discordant. The overarching goal in defining the elements of each domain was to prioritize specificity in attributing activity to neurosarcoidosis while minimizing the risk of misattribution. Each domain includes criteria for 4 categories of disease status: “improved,” “stable,” “worsened,” or “resolved.” The overall disease status of activity or inactivity is then determined by reconciling the independent domains into a summary disease activity assessment, communicated with additional clinical context, including immune treatment status and neuroanatomic localization. Sustained clinical stability of persistent neurologic symptoms or signs in the absence of MRI or CSF activity is considered in the summary assessment to reflect inactive disease with sustained neurologic deficits.

Summary

A systematic review highlighted substantial heterogeneity in how CNS neurosarcoidosis disease activity is measured, with many studies lacking formal definitions and often reporting activity only within isolated domains. This proposed framework integrates multiple domains to comprehensively characterize CNS neurosarcoidosis disease activity. With further validation, this approach has the potential to standardize research practices and improve clinical reasoning and communication.

MORE ONLINE

Supplementary Material

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Introduction

Sarcoidosis is a multisystem, inflammatory disease characterized by granulomatous inflammation.¹ Involvement of the nervous system by the sarcoidosis inflammatory process—neurosarcoidosis—affects between 5% and 30% of people with sarcoidosis. Neurosarcoidosis can affect nearly all peripheral and CNS compartments.^{2–11} The variability of neuroanatomical involvement in neurosarcoidosis results in a broad spectrum of clinical presentations, phenotypes, temporal evolutions, and outcomes.

Consensus criteria for the diagnosis of neurosarcoidosis provide a helpful framework for diagnostic reasoning and communicating degrees of diagnostic confidence in clinical care and research.¹² Although biopsy evidence of sarcoidosis in an affected organ system increases diagnostic confidence, even typical histopathologic findings are not entirely specific and require rigorous clinical correlation and exclusion of other potential causes.

Given the absence of a single test or biomarker to support neurosarcoidosis diagnosis or measure disease activity, the longitudinal care of people with neurosarcoidosis relies on dynamically integrating data from various domains, including clinical, imaging, and laboratory measures, into a summary impression. However, there is currently no standardized or consensus-based clinical framework for characterizing and monitoring disease activity in CNS neurosarcoidosis once the diagnosis has been established, which leads to substantial variability in clinical practice and inconsistency in research methods. Many observational studies focus on outcomes within single domains (e.g., clinical, imaging, or laboratory measures), frequently without formalized criteria. No randomized controlled trials have been conducted with neurologic-focused outcomes in systemic sarcoidosis or have there been any for neurosarcoidosis. Consequently, CNS neurosarcoidosis treatment strategies rely on observational data or post hoc analyses.^{9,13–16} The absence of a harmonized framework for assessing disease activity in CNS neurosarcoidosis perpetuates reliance on lower levels of evidence, limits comparisons across cohorts, and hinders formal meta-analysis, which is particularly important for a rare disease.

An NIH-funded, multicenter, prospective cohort study of people with CNS neurosarcoidosis presented an opportunity to develop a framework for longitudinal, disease activity monitoring.¹⁷ We conducted a systematic review to compile definitions of disease activity described from the peer-reviewed literature and developed integrated definitions for neurosarcoidosis activity levels. Our proposed framework includes 3 data domains—clinical, imaging, and laboratory—and provides a standardized approach when domain results are discordant. This structure builds on previous efforts to standardize disease activity definitions in multiple sclerosis (MS) clinical trials^{18–21} and systemic sarcoidosis observational studies,^{22,23} addressing the need to

integrate multiple data streams for clinical management without reliance on a single biomarker.

Similar to the diagnostic criteria used for neurosarcoidosis, this proposed framework for disease activity in neurosarcoidosis relies on clinical expertise, clinical context, and clinician judgment. The framework, therefore, is designed to augment routine clinical care and support more standardized communication pertinent to a complex neurologic disease.

Methods

Systematic Review

Search Strategy

The systematic review proposal was registered through PROSPERO (ID: 533857).²⁴ We searched the PubMed and Embase databases for English language articles published from January 2000 to April 2024 using the terms “Neurosarcoidosis,” “CNS Sarcoidosis,” “Central Nervous System Sarcoidosis,” or “Muscular Sarcoidosis” in the title or abstract. Abstracts, reviews, and case reports were filtered using the search strategy. Articles were included in the systematic review if they were (1) published in the English language, (2) included at least 1 longitudinal monitoring metric, (3) included a sample size of at least 5 or more patients with probable or definite neurosarcoidosis, (4) included an analytical focus that was primarily neurosarcoidosis as opposed to another disease, and (5) was an original research article. The complete search strategy is available in eFigure 1. Articles were reviewed and data extracted for study design, country of the study site, dates of data collection, basic population descriptions (sample size, average age, race, proportion of biopsy-proven sarcoidosis, and neurosarcoidosis cases), and definitions of clinical, imaging, CSF, serum, and patient-reported metrics used for longitudinal monitoring of neurosarcoidosis. All studies in the review were included in summary tables, and if there were missing data, the number of studies with missing data was reported.

Article Screening and Extraction

Two reviewers (G.W. and J.M.G.) independently examined titles, abstracts, and papers to determine inclusion in the systematic review. One reviewer (G.W.) performed data abstraction and bias assessment. The assessment of bias was performed using the Cochrane Risk of Bias Tool.²⁵ In addition, the authors assessed each study for the quality of evidence based on the Oxford Level of Evidence.²⁶ Disagreements were resolved by rereview and discussion by the reviewers. Title and abstract screening, full-text screening, bias assessment, and data extraction were performed using Covidence, a web-based collaboration software platform that streamlines the production of systematic and other literature reviews.²⁷ Results presented according to Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.

Proposed Disease Activity Definitions

Integrating results from the systematic review, we developed a CNS neurosarcoidosis-specific, multidomain framework for assessing disease activity, including definitions of activity, inactivity, remission, and relapse through nonblinded, open discussions among the authors. This multicenter, NIH-funded, prospective neurosarcoidosis study brought together experts at the University of California San Francisco (UCSF), Johns Hopkins, and Cedars-Sinai. The process involved team meetings, integration of data from the systematic review, and expert opinion to develop and refine a framework that ultimately reflected group consensus. This framework is modeled after the multidomain approach used to characterize disease activity in MS research.¹⁸

Standard Protocol Approvals, Registrations, and Patient Consents

This analysis did not require ethical approval as it only used published, aggregate patient-level, deidentified data (systematic review).

Data Availability

All data acquired through this study are published in supplemental data sections and will be available.

Results

Systematic Review

Of the 387 articles in the initial search, 271 were excluded based on title review: 88 case reports or case series with fewer than 5 patients, 47 review articles without primary data, 48 without a longitudinal monitoring metric, 76 in which neurosarcoidosis patients were not the analytic focus, 6 pediatric populations, 3 non English, and 3 author-response letters. This left 116 studies for full-text review, of which 49 were excluded: 5 case reports, 31 without a longitudinal monitoring metric, 2 non-English language full text, 7 in which neurosarcoidosis patients were not the analytic focus, 2 systematic reviews, and 2 review articles without original data. The systematic reviews were excluded because all references they cited were already included in our review, except for 1 paper, which was added to our full-text screening. Ultimately, 67 articles were included, and details of the parameters used to monitor neurosarcoidosis disease activity were extracted (Figure 1).

Most studies were retrospective cohort studies ($n = 59$, 88%), including 12 multicenter studies and 2 interventional retrospective cohort studies. The sample sizes ranged from 5 to 234, with a mean of 40 patients. Most studies were published between 2011 and 2020 ($n = 34$, 51%) and in the United States ($n = 31$, 46%) or France ($n = 16$, 24%). Approximately half of the studies ($n = 36$, 54%) focused on specific patient populations, such as those with spinal cord sarcoidosis ($n = 7$, 11%), a treatment-based cohort ($n = 12$, 18%), or neuro-ophthalmologic sarcoidosis ($n = 4$, 6%). The

proportion of patients with probable neurosarcoidosis (systemic biopsy proven) ranged from 17% to 100%, whereas definitive neurosarcoidosis (nervous system biopsy confirmed) ranged from 0% to 83%. Study characteristics, including baseline demographics and monitoring metrics, are summarized in Table 1, with additional study-specific and methodologic details in eTable 1. According to PRISMA guidelines, examples of studies considered for inclusion but ultimately excluded and the justification for exclusion are presented in eAppendix.

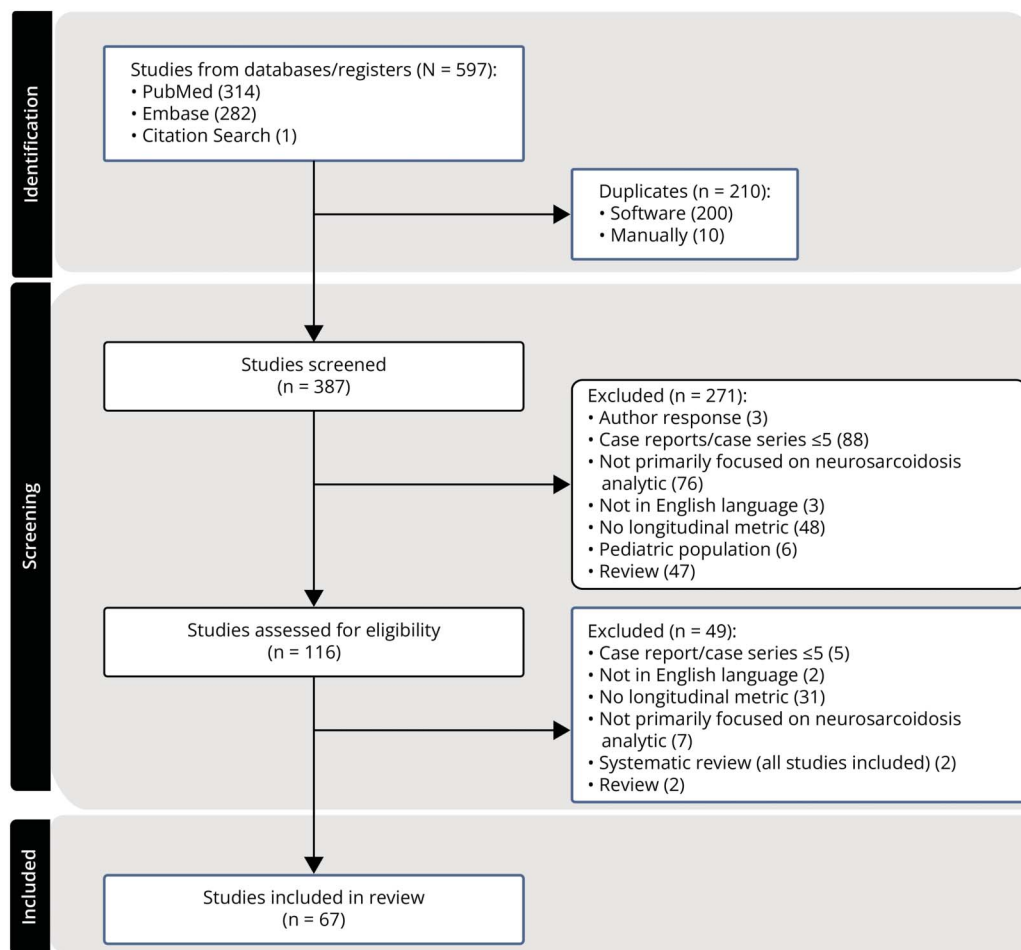
Of the 67 studies included in the systematic review analysis, most reported an independent clinical monitoring outcome ($n = 59$, 88%), of which 24 (41%) used the modified Rankin scale, followed by 12 (20%) relying on the treating clinician's assessment of clinical response without formal criteria. Twenty-seven percent of the studies with a clinical outcome metric did not formally define the clinical monitoring metric, often using terms such as "improved, stable, or worsening" without specific criteria definitions (Table 1). Detailed narrative descriptions of study-specific clinical definitions are reported in eTable 2.

An independent imaging assessment metric was included in 29 studies (43%). The most common imaging metric was an abnormal enhancement on T1 postcontrast MRIs after the administration of gadolinium ($n = 11$, 38%), followed by lesion size ($n = 8$, 28%) and lesion quantity ($n = 7$, 24%) on MRI T2 sequences. In 31% of studies with imaging assessment metrics, these criteria were not formally defined, with 17% of studies characterizing changes as improved, stable, or worsening without specific definitions. A summary of imaging assessment metrics is included in Table 1. The full descriptions of study-specific imaging definitions are reported in eTable 2.

Only 7 studies (10%) included an independent CSF longitudinal assessment metric. Of these, only 2 (3%) specified CSF assessment criteria, measuring CSF white blood cells, glucose, and protein. In 2 studies (30%), the CSF analysis metric was not formally defined. Only 4 studies included an independent serum assessment metric, which involved markers such as serum angiotensin-converting enzyme and soluble interleukin-2 receptor (sIL-2r). A complete list of CSF and serum monitoring metrics is provided in Table 1.

Although most studies included a single disease activity outcome metric (i.e., clinical, imaging, CSF, or serum measurement), only 11 studies (16%) included a multidomain assessment metric, integrating both clinical and imaging metrics. Detailed descriptions of study-specific combined evaluation metrics are reported in eTable 2.

Among the studies that included a clinical definition for relapse, the terminology differed, with terms such as "remission," "complete remission," "partial remission," "improving," and "stable," "stability," "maintenance," and

Figure 1 Consort Diagram for Systematic Review

“deterioration,” “worsening,” and “relapse” used interchangeably. Most studies did not formally define these terms; complete descriptions of those that did are reported in eTable 2.

Analysis of the assessment of bias for studies included in the systematic review is reported in the eAppendix, specifically eFigure 2 and eTable 3.

Proposed Domain-Specific Disease Activity Definitions in Neurosarcoidosis

Based on a comprehensive and systematic literature review, we developed a framework to define CNS disease activity across 3 domains; (1) clinical, (2) imaging, and (3) laboratory. The domain-specific definitions are summarized in Figure 2. Each domain is assessed independently of the others and then integrated into a summary assessment of disease activity, communicated with additional clinical context, including immune treatment status and neuroanatomic localization. Within the clinical and imaging domains, the “active” disease category is divided into 3 ordinal sublevels: “improved,” “stable,” and “worsened.” All domains include

the inactive category of “resolved.” Sustained stable or improved clinical symptoms or signs in the absence of imaging or laboratory activity are considered in the summary assessment to reflect inactive disease with sustained neurologic deficits. When some components informing activity assessment are unavailable clinically (i.e., associated neurologic examination, noncontrast imaging, or specific laboratory studies), disease status is determined from the available components when able and communicated in the context of available data.

The overarching goal for defining which elements to include in the framework within each domain is to prioritize specificity for neurosarcoidosis while minimizing the risk of misattribution.

Clinical Domain

Clinical domain assessments used for this framework must occur in the absence of fever or delirium and after evaluation for other potential confounding conditions (e.g., infectious meningitis, acute stroke). The assessment should occur within the context of relatively stable systemic disease

Table 1 Characteristics of Studies Included in the Systematic Review (N = 67)

Study characteristic	N (%)
Study design	
Case series	8 (12)
Interventional retrospective cohort study	2 (3)
Multicenter retrospective cohort study	12 (18)
Retrospective cohort study	45 (67)
Prospective cohort study	3 (4)
Publication year	
2000–2010	16 (24)
2011–2020	34 (51)
After 2020	21 (31)
Sample size mean (SD)	40 (60)
Study site location^a	
France	16 (24)
Germany	4 (6)
The Netherlands	6 (9)
The United States	31 (46)
Length of retrospective review (y, mean [SD]) (n = 59)	12 (18)
Retrospective review start y	
Before 1970	2 (3)
1970–1979	2 (3)
1980–1989	3 (4)
1990–1999	17 (25)
2000–2009	13 (19)
After 2010	18 (27)
Missing	13 (19)
Special patient population	
No specification	35 (52)
Cranial skull based	1 (1)
Cauda equina	1 (1)
Central DI/hypogonadism/hypothalamus-pituitary	2 (3)
Myelitis/myelopathy/spinal cord sarcoidosis	7 (10)
Treatment based cohort	13 (19)
Facial nerve	1 (1)
Hydrocephalus	2 (3)
Ischemic stroke	1 (1)

Table 1 Characteristics of Studies Included in the Systematic Review (N = 67) (continued)

Study characteristic	N (%)
Isolated neurosarcoidosis	1 (1)
Muscular sarcoidosis	2 (3)
Neuro-ophthalmic sarcoidosis	4 (6)
Pachymeningitis	1 (1)
Small figure neuropathy	1 (1)
Tumefactive brain lesions	1 (1)
Mean age (y) (n = 65) mean (SD)	43 (6)
Percent female sex (n = 70) mean (SD)	54 (13)
Percent White race ^b (n = 37) mean (SD)	57 (31)
Percent Black-African American race ^b (n = 37) mean (SD)	44 (29)
Percent with probable neurosarcoidosis ^c (n = 53) mean (SD)	67 (22)
Percent definitive neurosarcoidosis ^c (n = 49) mean (SD)	22 (24)
Median follow-up time (months) (n = 53) mean (SD)	37 (28)
Domains of longitudinal monitoring metric^d	
Independent clinical monitoring	59 (88)
Independent radiological monitoring	29 (43)
Independent CSF monitoring	7 (10)
Independent serum monitoring	4 (6)
Combination (multidomain) monitoring	11 (16)
Clinical monitoring metrics (n = 54)	
Modified Rankin Scale	24 (41)
Not formally defined	16 (27)
Treating clinical discretion	12 (20)
Visual acuity	6 (10)
Study specific clinical definition	4 (7)
Symptoms	4 (7)
ePOST	3 (5)
Expanded Disability Status Scale (EDSS)	3 (5)
Ambulation status	1 (2)
Body mass index	1 (2)
EDMUS Grading Scale (EGS)	1 (2)
House Brackmann Score	1 (2)
Menstrual cycle	1 (2)
Modified Oxford Handicap Scale (MOHS)	1 (2)

Continued

Table 1 Characteristics of Studies Included in the Systematic Review (N = 67) (*continued*)

Study characteristic	N (%)
Seizure control	1 (2)
Disease steps in multiple sclerosis (STEPS)	1 (2)
Radiologic monitoring metrics (n = 29)	
Gadolinium enhancement	11 (38)
Not formally defined	9 (31)
T2 lesion size	8 (28)
T2 lesion quantity	7 (24)
"Improvement, stable, or worsening"	5 (17)
Hydrocephalus	1 (3)
PET avidity	1 (3)
Ischemic stroke	1 (3)
Apparent diffusion coefficient values	1 (3)
"Abnormalities related to neurosarcoidosis"	1 (3)
Complete or near complete resolution	1 (3)
CSF monitoring metrics (n = 7)	
White blood cell count; glucose; protein	2 (30)
White blood cell count	1 (17)
Interleukin 6	1 (17)
Pleocytosis	1 (17)
Not formally defined	2 (30)
Serum metrics^d	
Angiotensin converting enzyme	2 (3)
Soluble interleukin-2 receptor	2 (3)
Pituitary hormones	1 (1)
Erythrocyte sedimentation rate	1 (1)

^a Highest 4 countries.^b Only White and Black Race presented because only 2 studies reported races other than White or Black.^c Based on criteria defined in the individual paper.^d Percentages do not sum to 100, as papers can have multiple independent monitoring metrics.

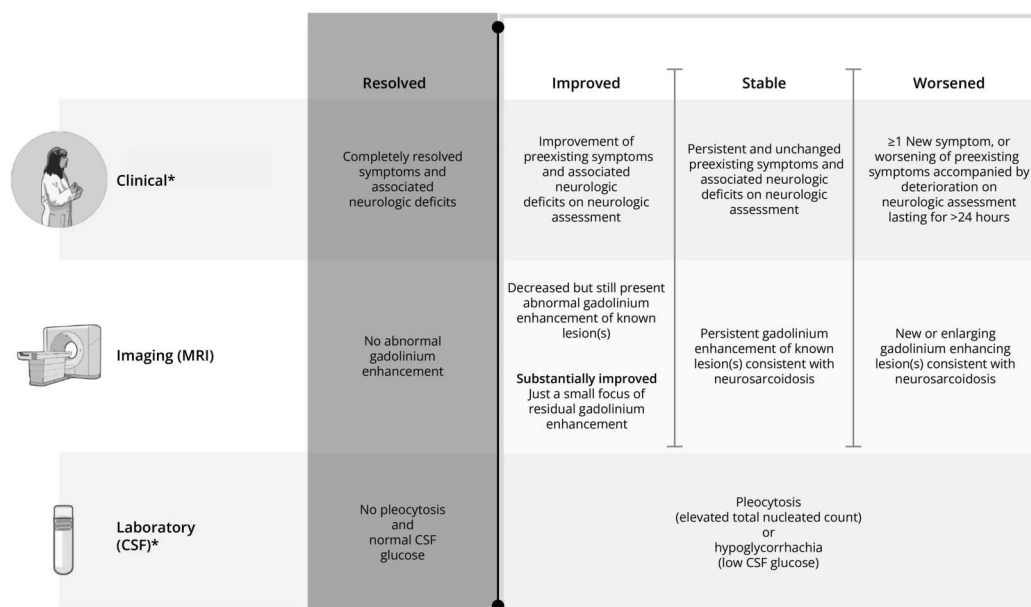
processes that have the potential to confound the evaluation (e.g., congestive heart failure, chronic hip pain affecting mobility unrelated to neurosarcoidosis), as determined by the treating or research-evaluating clinician. The clinical domain for disease activity is meant to reflect clinical disease that would meet consensus criteria for CNS neurosarcoidosis symptoms. If symptoms are from para (neuro) sarcoidosis, then a separate clinical distinction can and should be made to provide a differential diagnosis and assessment for these symptoms. Disease activity within all

domains is defined by the categories of "resolved," "improved," "stable," or "worsened." "Improved disease" is defined as an improvement or absence of preexisting symptoms and associated deficits on neurologic assessment. "Stable disease" is defined as unchanged preexisting symptoms and associated deficits on neurologic assessment. "Worsened disease" is defined as 1 or more new symptoms or worsening of preexisting symptoms accompanied by neurologic assessment deterioration lasting more than 24 hours. Neurologic assessments can include the neurologic examination but also other validated symptom-specific assessments per local practice. Headache can be a key symptom of many neurosarcoidosis manifestations, including meningitis and hydrocephalus. Headache may also be from comorbid disease, including primary headache disorders, which are common in general and in people with sarcoidosis.²⁸ To include headache reliably as an element of disease status in neurosarcoidosis, it must be supported by other evidence (i.e., MRI evidence of inflammation, hydrocephalus, or CSF abnormalities) and because it cannot stand independently it is excluded from the clinical domain in this framework. In the absence of a clear alternative explanation (i.e., epilepsy syndrome, a mass that is not thought to be sarcoidosis [e.g., meningioma], toxic or metabolic triggers such as alcohol withdrawal, hypoglycemia, etc.), new seizures can be considered as clinical worsening without lasting 24 hour or requiring neurologic deficits on examination if thought attributable to neurosarcoidosis. In clinical practice and in clinical research, the distinction between new symptoms or worsening of established symptoms should be included as a modifier, as per usual clinical practice, by recording whether the clinical context includes new or worsening symptoms.²⁹

Clinical "resolved disease" is met when previously present symptoms attributed to neurosarcoidosis are no longer symptomatic.

Imaging (MRI) Domain

All assessments in the imaging domain rely on MRI with and without the administration of gadolinium when available. This involves imaging of the relevant affected neuroanatomic structures based on neurologic localization (i.e., for myelopathy symptoms, an MRI of the spine is obtained; for headache symptoms, an MRI of the brain is obtained). When MRI is not available or contraindicated, other imaging modalities, such as a head CT, may be considered, although they are generally less sensitive or specific. The decision to use alternative imaging techniques will depend on the context; for example, including non-MRI imaging assessment may still be helpful in clinical care in select cases, but could risk introducing bias in some research analyses. In the imaging domain, "improved disease" is defined as decreased but persistent abnormal gadolinium enhancement on post-contrast T1-weighted sequences. Within "improved disease," there is a distinction for "substantially improved" which is defined as areas of significantly improved gadolinium enhancement on T1-weighted sequences in which the

Figure 2 Proposed Domain-Specific Disease Activity Definitions

A graphic summary of the cross-sectional definitions for each level of disease activity within each domain. *In the absence of fever and after laboratory evaluation for other explanations (e.g., infectious meningitis, seizure, or stroke) as clinically indicated. Exception: Headache syndromes are excluded from clinical domain analysis as they require supportive data from imaging and laboratory domains. Exploratory CSF outcomes include soluble IL-2 receptor, IgG Index, unique CSF oligoclonal bands, elevated protein greater than 2 times the upper limit of normal. These should be included when available but are not required. Hypoglycorrhachia is typically defined as less than 45 mg per deciliters but correlation of CSF glucose with a serum glucose is essential, as a normal CSF to serum glucose ratio is typically 0.5–0.6 or greater, although slightly lower ratios may be observed in patients with serum hyperglycemia.

abnormal contrast enhancement is nearly completely resolved except for a small focus of residual enhancement. “Stable disease” is defined as persistent abnormal gadolinium-enhancement of the known lesion(s). “Worsened disease” is defined as a new or enlarging gadolinium-enhancing lesion(s). “Resolved disease” is defined as the absence of abnormal gadolinium enhancement. The degree of contrast enhancement is conceptualized as a continuum from (the most) worsened, stable, improved, substantially improved, and resolved (no abnormal enhancement). The goal of the category of “substantially improved” is to allow for the distinction between a persistent, small focus of residual gadolinium enhancement on postcontrast T1 MRI sequences that may not ultimately be a marker of active inflammation, as compared with improving, but more florid, widespread presence of gadolinium enhancement.

Leptomeningeal nonsuppression hyperintensity on post-contrast fluid-attenuated inversion recovery (FLAIR) is considered an exploratory marker of activity in the imaging domain when it is plausibly related to the leptomeningitis of neurosarcoidosis, after the exclusion of other potential causes (which may require CSF sampling).³⁰ In an effort to design a simpler, standardized framework, we have elected not to include isolated changes in T2/FLAIR lesion parameters as a category in the imaging domain. The development of new T2/FLAIR lesions in individuals with age, microvascular disease, and other risk factors is common and less specific. The rationale for the exclusion of these lesions focuses on avoiding misattribution of disease activity based on non-

neurosarcoidosis-specific lesions. Future iterations of this framework may allow for neurosarcoidosis specific T2/FLAIR lesions if there is reasonable specificity and adjudication. Imaging inactivity is met when no abnormal gadolinium-enhancing lesions are attributable to neurosarcoidosis on postcontrast T1-weighted MRI sequences.

CSF Laboratory Domain

All assessments in the laboratory domain are based on CSF analysis and the performing laboratory’s reference ranges. This domain includes both core criteria and exploratory outcomes, but there are no subcategories for disease activity. “Active disease” is defined as any of the following metrics: pleocytosis (elevated CSF total nucleated cell count, typically defined as more than 5 per cubic millimeter of CSF) or hypoglycorrhachia (low CSF glucose). Hypoglycorrhachia is typically defined as less than 45 mg per deciliter. Correlation of CSF glucose with a serum glucose is also essential clinically, as a normal CSF to serum glucose ratio is typically 0.5–0.6 or greater, although slightly lower ratios may be observed in patients with serum hyperglycemia without CNS infection or inflammation (with ratios as low as 0.4 for serum glucose up to 149 mg/dL and 0.3 for serum glucose >150 mg/dL, in which case correlation with other CSF and clinical metrics is helpful for interpretation and clinical contextualization).^{31–33} Elevated IgG index, elevated total protein (especially if it exceeds twice the upper limit of normal), and elevated sIL-2r, are also supportive of active inflammation in neurosarcoidosis but are considered as

exploratory outcomes within the laboratory domain to be validated for core inclusion with further study. Oligoclonal bands were not included in the current framework given the need for more research to determine the clinical relevance of the presence and persistence of OCBs for disease activity monitoring in neurosarcoidosis. As new serum or CSF biomarkers of disease activity are discovered and validated, they should be incorporated into future updates of this domain. Such as the clinical domain, the interpretation of CSF results must consider potential confounding factors (e.g., infectious meningitis in the context of immunosuppression) as determined by the treating or research-evaluating clinician. Detailed lists of differential diagnoses in neurosarcoidosis have been outlined previously,¹² and it remains important to exclude CNS infection, especially when the patient is treated with immune therapies, and neoplasm, especially lymphoma.

Laboratory inactive disease criteria are met when the core criteria (pleocytosis, hypoglycorrhachia) normalize. If using exploratory outcomes (such as CSF oligoclonal bands, IgG index, total protein levels normalized below twice the upper limit of normal if previously elevated, or normalized sIL-2r), inactivity is determined by normalization of these outcomes.

Proposed Overall Disease Activity Definitions

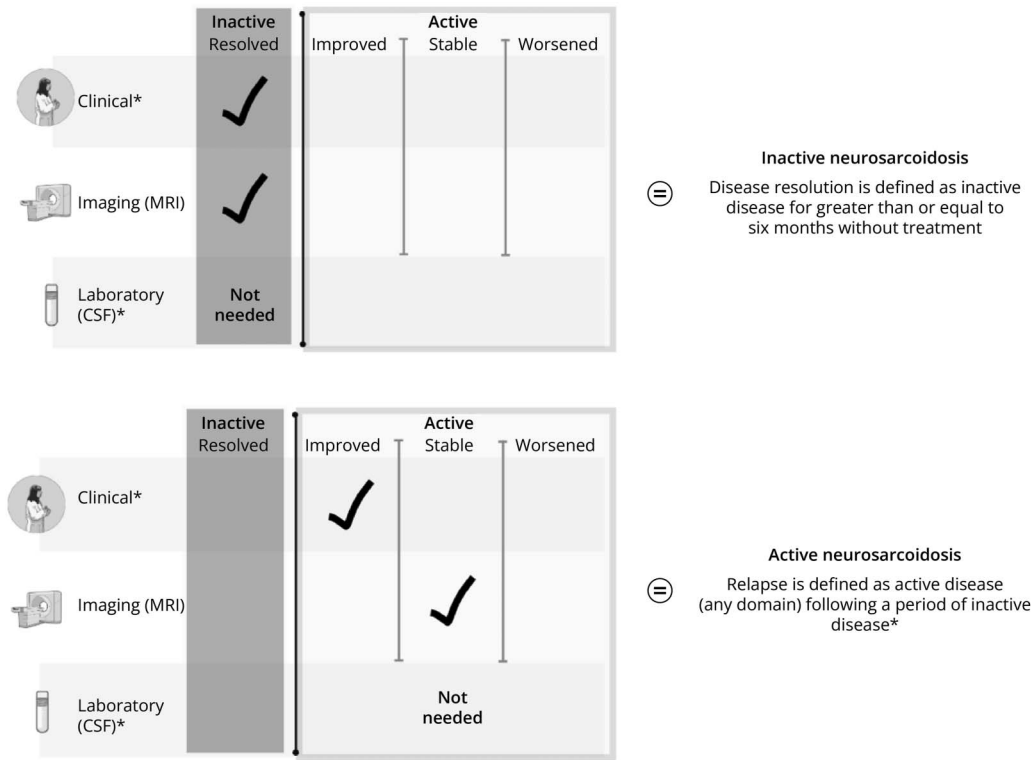
Each domain’s assessment occurs independently and cross-sectionally, allowing for discordant disease status assessment

across domains. Therefore, assessment of overall disease activity status requires reconciliation of the independent domains. Because of the invasive nature of the laboratory (CSF) domain, it is not as routinely obtained as clinical and imaging data. Therefore, if the clinical and imaging domains are concordant, no further reconciliation with the laboratory domain is required. This framework for overall disease activity, using the domain-specific definitions above, is outlined in Figures 3 and 4. Complete permutations of cross-sectional disease status can be found in eFigure 3.

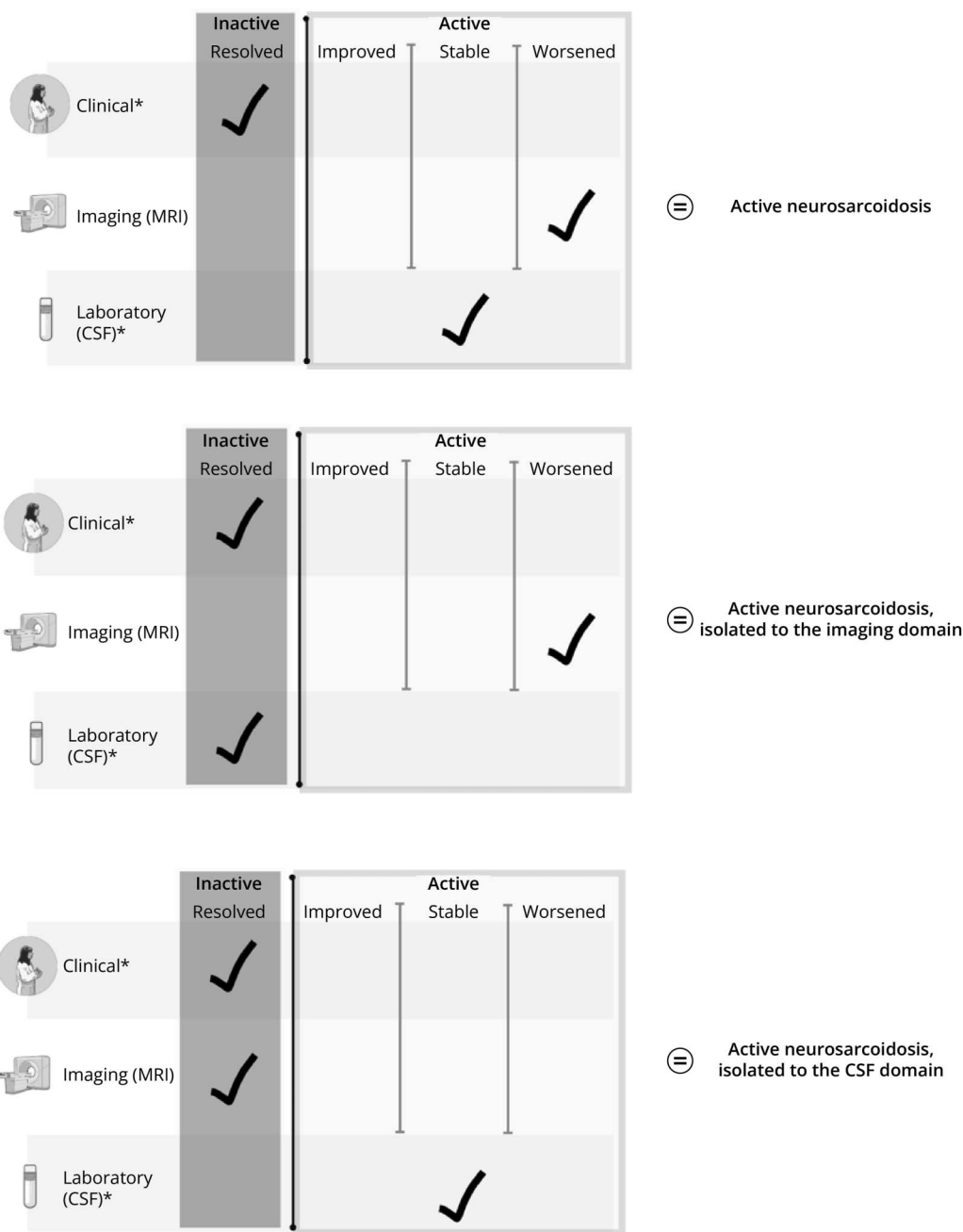
If clinical and imaging disease activity determinations are discordant, laboratory evaluation can help determine overall disease status. When all 3 domains are evaluated, if 2 of the 3 reflect active disease, then the overall disease status is considered active. Conversely, if 2 domains indicate inactive disease, then the disease status reflects isolated disease activity in the domain that remains active. If laboratory domain data is helpful but not obtained, then the disease status should be reported as is (e.g., “imaging active and clinically inactive”) to reflect this discordance. This framework for overall disease activity, using the domain-specific definitions above, is outlined in Figure 4.

Cross-sectional assessments do not include embedded time components, but summary disease status may be contextualized within the context of time (inactive for X number of

Figure 3 Concordant Clinical and Imaging Cross-Sectional Assessments



A graphic summary of disease activity determination with concordance between clinical and imaging domains.

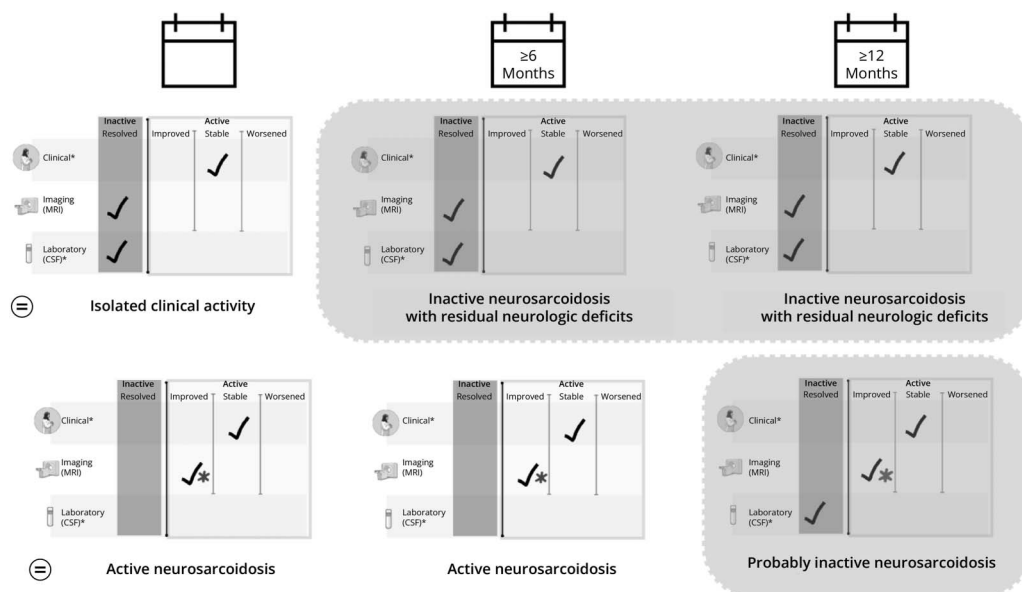
Figure 4 Discordant Clinical and Imaging Cross-Sectional Assessments

A graphic summary of disease activity determination with discordance between clinical and imaging domains.

months since last assessment) as per standard clinical reasoning. Overall disease activity determination, however, incorporates longitudinal assessments across time. For example, without a longitudinal assessment, if a patient demonstrated clinical activity with persistence of neurologic deficits (even if stable for many years), they could never be deemed inactive. The inclusion of a longitudinal assessment allows for a patient's disease activity status to become inactive if there are persistent residual clinical deficits in the context of sustained inactivity in the other domains (i.e., Isolated Clinical Disease Activity) for at least 6 months.

This reflects the clinical observation that neurosarcoidosis can sometimes lead to neurologic injury with sustained neurologic deficits that persist or incompletely heal, even when the disease process and abnormal inflammation of neurosarcoidosis otherwise remit (Figure 5). Another example incorporating longitudinal assessments in overall disease activity is for a patient with stable clinical disease, substantially improved radiologic disease, and resolved laboratory disease activity for at least 1 year. In this circumstance, the overall disease status transitions to "probably inactive disease" (Figure 5).

Figure 5 Longitudinal, Summative Disease Assessments



A graphic summary of overall disease activity determination incorporating multiple cross-sectional assessments across time. *Substantially Improved Imaging Activity.

Overall disease resolution is defined as inactive disease lasting at least 6 months without any immune disease-modifying therapy. Disease relapse is defined as meeting active disease criteria in any domain after a period of overall inactivity. The terms “resolution” and “relapse” were chosen to align with terminology used in the systemic sarcoidosis literature.^{22,23,34} The term “relapse” implies that the disease must have entirely resolved (i.e., inactive disease status) before it can recur. Therefore, if a patient experiences worsening disease activity without previous inactivity, this is classified as a worsening of disease activity rather than a “relapse.” Finally, a comprehensive assessment of disease activity should include descriptions of the pertinent affected neuroanatomic site and associated neurologic syndrome. For example, a patient with a spinal cord lesion could be described as having myelitis attributed to active neurosarcoidosis or a history of myelitis attributed to a now inactive neurosarcoidosis, depending on the context. Similarly, in a patient with multifocal neurosarcoidosis (i.e., neurosarcoidosis affecting multiple areas of the CNS), each neuroanatomic location should be assessed. For example, within the same patient, pachymeningitis could meet the overall criteria for inactive disease whereas the myelitis could meet the criteria for active disease, and overall, the patient would have active disease. Common clinical vignettes with disease status designations are presented in eAppendix.

Discussion

Our systematic review demonstrated the heterogeneity in approaches and the predominance of single-domain (clinical, imaging, etc.) longitudinal disease monitoring assessments for CNS neurosarcoidosis. Specifically, and remarkably, only

11 of 71 studies included monitoring assessments that combined even 2 domains of disease activity. These longitudinal assessments were often poorly defined and frequently relied on scales initially designed for stroke and MS populations.^{35,36} This systematic review is the first of its kind and underscores the urgent need for a harmonized framework that integrates multidomain parameters and more formalized definitions of disease activity in neurosarcoidosis.

A limitation of this systematic review is the focus on CNS manifestations of neurosarcoidosis and the lack of quantitative analysis regarding the strength of the assessment metrics. Because the focus was on aggregating existing definitions, we did not assess the utility of each metric or its associations with functional outcomes. The review process had an English language restriction and was conducted by only 2 reviewers, although there were minimal disagreements.

To improve the rigor, standardization, and interoperability of future observational and interventional studies in CNS neurosarcoidosis, we developed multidomain definitions for disease activity with a framework for monitoring disease activity. Rigorous validation studies are needed to confirm the practical utility of this framework and its associations with predefined, clinically meaningful outcomes. Our proposed model may not directly correlate with the functional status or quality of life of patients with neurosarcoidosis. Important areas for validation include the subcategory of “substantially improved” within the imaging domain. This more specific imaging subcategory is designed to enable a more nuanced description of the imaging while allowing for the hypothesis and clinical observation that, particularly in certain

neuroanatomic locations, focal gadolinium enhancement may not necessarily reflect active CNS inflammation and can sometimes result from other pathophysiologic factors such as fibrosis, prior biopsy, ischemia, vascular abnormalities, or other related chronic noninflammatory disruptions of the blood-brain barrier.

Although some patients with neurosarcoidosis will meet inactivity criteria in all domains, some may have residual persistent neurologic deficits from nervous system injury even when the disease process and abnormal inflammation of neurosarcoidosis remit. The “inactive disease with residual clinical deficits” designation allows for efficient and nuanced communication between clinicians, not just about overall disease status but also neurologic status, while explicitly highlighting the domain that requires continued monitoring and attention. For example, a patient who developed vision loss in the setting of sarcoidosis-associated optic neuropathy could, after 6 months of stable, persistent decreased visual acuity in the sustained absence of abnormal gadolinium enhancement or pleocytosis, be assigned a status of “inactive neurosarcoidosis optic neuropathy with residual clinical deficits.” The cutoff of 6 months of stability is not meant to be prescriptive or prognostic but simply an attempt to select a reasonable period of time for clinical stability to inform the overall assessment of disease activity.

Similarly, by integrating multiple domains and time, the overall status of “probably inactive disease” captures the pathologic uncertainty of the residual focus, particularly if it may not reflect active inflammation. The cutoff of 1 year is again not considered prescriptive or prognostic, but rather to allow for a reasonable amount of time for the opportunity for resolution of imaging and clinical disease activity before a presumptive disease status change.

This framework is designed to improve characterization, assessment, and communication of disease activity in neurosarcoidosis and not to inform prognostication or dictate decisions about treatment, and additional validation studies will be essential for assessing their utility.

Our systematic review highlights the lack of quality evidence or even common language for longitudinal follow-up in neurosarcoidosis, which was the impetus for this proposed framework for defining disease activity and why this approach has the potential to be particularly impactful for the field with further validation. These definitions are meant to provide a call to action for a shared language and a common framework for conceptualizing disease activity in neurosarcoidosis. Future efforts might entail developing functional and quality-of-life metrics for the neurosarcoidosis population. This may involve repurposing scales designed for other disorders, creating unique and novel scales developed for neurosarcoidosis, and using data from digital “wearables” and life space activity analyses.³⁷ For this initial framework, the clinical domain does not specify or favor specific scales or

quantitative metrics beyond the standard neurologic examination and locally used validated symptom questionnaires or rating scales. This approach allows for flexibility and reflects clinical practice, but will require selection of specific scales and adjudication for translation to research and clinical trials.

In summary, this proposed framework integrates multiple domains to precisely categorize disease activity in neurosarcoidosis and aims to standardize research while enhancing clinical reasoning and communication in routine patient care. There are no data to support their use in guiding treatment decisions or prognostic features at this time. We anticipate that this framework will evolve through validation studies and as more specific and reliable biomarkers of disease activity in neurosarcoidosis are identified.

Author Contributions

G. Waldrop: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. S.C. Dominguez Penuela: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. P. Barreras: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. E. Kinder: drafting/revision of the manuscript for content, including medical writing for content. K.C. Fitzgerald: drafting/revision of the manuscript for content, including medical writing for content. H.B. Larman: drafting/revision of the manuscript for content, including medical writing for content. S. Pleasure: study concept or design. M.R. Wilson: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. B.J. Stern: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. C.A. Pardo: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. J.M. Gelfand: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

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